Roll No: _____

Thapar Institute of Engineering and Technology
Department of Computer Science and Engineering
Auxiliary Examination

BE CSE Third Year
Course Name: Quantum Computing
Time: 3 hours (5:30 pm-8:30 pm)
Marks : 100

Course Code: UCS619
Date: February 13, 2025
Name of Faculty: Dr Samya Muhuri

Note: Attempt all questions with proper justification and working. State your assumptions (if any) and show your working clearly.

1. (a) A system in the state

$$|\psi\rangle = \frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

What is the probability that measurement finds system in the state $\phi = |01\rangle$? Find using Born rule. **(6)**

(b) What is the probability that measurement finds the second qubit in the state $|1\rangle$? What is the state of the system after measurement? **(6)**

2. (a) The matrix product

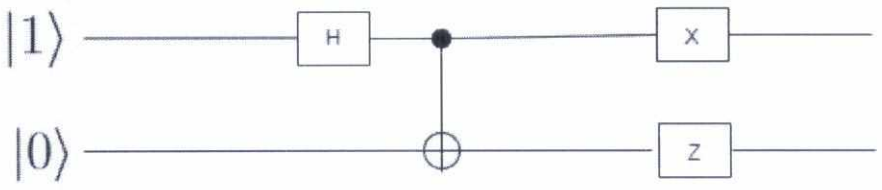
$$|a\rangle \otimes |b\rangle = \frac{1}{\sqrt{10}}\begin{pmatrix} -2 \\ 1 \\ 2 \\ -1 \end{pmatrix}$$

and

$$|a\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Find $|b\rangle$. **(4)**

(b) What is the output at each step of the following circuit?



(4)

(c) Show that the Toffoli gate can act as AND Gate. **(4)**

3. (a) The standard way to write the state of a qubit using the bloch sphere is

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

show that on the y-axis of the Bloch Sphere $y_+ = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$

(6)

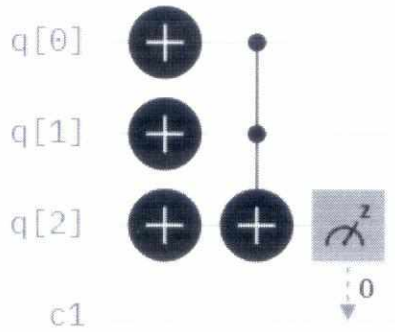
(b) Using the matrix representation of the Hadamard gate, write down the matrix $(H \otimes H)(|0\rangle \otimes |1\rangle)$. Show that this is equivalent to

$$|\psi\rangle = (\frac{|0\rangle + |1\rangle}{\sqrt{2}})(\frac{|0\rangle - |1\rangle}{\sqrt{2}})$$

(6)

4. (a) Write down the different steps for creating shifted key based on E91 protocol by considering the entangled pair $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. Explain it with the help of an example. **(6)**

(b) Consider OR ($\vee$) Gate as shown in Figure. Here $q[2] = q[0] \vee q[1]$.



Create a circuit for $(x_0 \wedge x_1) \wedge (x_1 \vee x_2)$. Here $\wedge$ represents AND Gate. x_0, x_1 and x_2 represents first, second and third qubits. You can use only three additional qubits. Draw only the quantum circuit. **(6)**

5. (a) A quantum state is in the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{e^{i\theta}}{\sqrt{2}}|1\rangle$$

Is the state normalized? Express the state in the $|+\rangle$, $|-\rangle$ basis.

(6)

(b) Proof No cloning theorem with two methods. **(6)**

6. (a) Define Quantum Fourier Transform (QFT) for an n-qubit quantum system. **(6)**

(b) Let $n = 1$ qubit, $N = 2$ dimensional space. Apply QFT on the state $\alpha|0\rangle + \beta|1\rangle$. **(6)**

7. (a) Write down a QASM program where you need to design a four qubits-controlled gate with three qubits acting as control qubits. If all three controlled qubits are equal to one, flip the target qubit. Use Two Taffoli gates and additional qubits if required. **(6)**

   (b) Construct a Quantum Circuit to implement the following four qubit Bell state $\frac{1}{\sqrt{2}}(|1001\rangle + |0110\rangle)$. Draw the quantum circuit and write down the QASM Code for the same. Give appropriate reasoning for your answers. **(6)**

8. Consider a function with two inputs such that $f(x) = 1$. Explicitly show that the Deutsch-Jozsa algorithm works in this case by generating the vector $|y\rangle = |00\rangle$ as the final output.

   (8)

9. Write down various steps of Shor's algorithm. Explain the procedure for factoring N=15. Please note that you need to explain the concept of Periodic function.

   (8)